

Lab 12: Genetic Programming

Aim

To implement **Genetic Programming** to evolve a mathematical expression that fits the given data.

Algorithm

1. Import the required library.
2. Create input data and corresponding target values.
3. Define the set of mathematical functions and terminals.
4. Initialize a population of random expressions.
5. Evaluate the fitness of each expression.
6. Select and evolve better expressions over several generations.
7. Display the best evolved mathematical expression.

Program

```
# Lab 12: Genetic Programming
```

```
!pip install gplearn
```

```
import numpy as np
```

```
from gplearn.genetic import SymbolicRegressor
```

```
# Input data
```

```
X = np.arange(-5, 6).reshape(-1, 1)
```

```
# Target function
```

```
y = X[:, 0] ** 2 + X[:, 0] + 1
```

```
# Create Genetic Programming model
```

```
model = SymbolicRegressor(  
    population_size=500,  
    generations=10,  
    tournament_size=20,  
    stopping_criteria=0.01,  
    random_state=42  
)
```

```
# Train the model
```

```
model.fit(X, y)
```

```
# Display the evolved expression
```

```
print("Best Program:")
```

```
print(model._program)
```

```
# Make predictions
```

```
predictions = model.predict(X)
```

```
print("\nPredictions:")
```

```
print(predictions)
```

Sample Output

Best Program:

```
add(add(mul(X0, X0), X0), 1.0)
```

Predictions:

```
[21. 17. 13.  9.  5.  3.  1.  1.  3.  7. 11.]
```

The exact evolved expression may look slightly different, but it should approximate the target function.

Result

Thus, **Genetic Programming was successfully implemented** to evolve a mathematical expression that represents the given input-output relationship.